Report Title

Design and Conceptual Implementation of Smart Kitchen Monitoring System based on Industry 4.0 technologies

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Abstract:

The issue of food safety and proper usage of food ingredients is a significant challenge in a Kitchen and restaurant environment. In this context, manual monitoring usually leads to food spoilage, waste, and even health risks. This report presents the design and analysis of a Smart Kitchen Monitoring System, which will leverage Industry 4.0 technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), Cloud Computing, Digital Twin concepts, and the concept of the Blockchain. Cloud computing enables scalable processing of data and making decisions in real-time, whereas blockchain allows verifiability of data integrity, transparency, and traceability of key food-related events. The functional prototype that focus on food spoilage detection through artificial sensor values and image samples is implemented to confirm the system viability. The results demonstrate that the proposed architecture can monitor the intelligent food safety, reduce food waste, and enhance the transparency in the work of the restaurant kitchen.

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1 Introduction

In the present day, there is a growing preference among people for eating at restaurants instead of preparing meals at home. In order to manage large numbers of customers and lower expenses, certain restaurants are engaging in unethical methods that include the use of food that is expired and have poor quality. These actions can be harmful and not only degrade the health of customers but also the restaurant's reputation and adherence to food safety standards. In the restaurant and kitchen industry, managing kitchen and food inventory brings both logistical and ethical challenges. The problem of food being wasted because of expiration is apparent in nearly all restaurants and requires attention. They appear to struggle with sorting their food products when these are nearing expiration. It is very important to prevent food from expiring and avoid fraudulent practices of using it that can harm customer health [2]. Food-borne illnesses affect an estimated 48 million Americans each year, or about one in six, and the food we eat often travels far before it arrives on our plates [9]. It is essential to keep an eye on the necessary conditions during this period to ensure that food storage remains consistent and suitable.

To tackle these issues, this project introduces a Smart Kitchen Monitoring System tailored for restaurant use and powered by Industry 4.0. It leverages digital twin technology, Internet of Things (IOT), blockchain technology, Cloud Storage and computing, and data management to intelligently manage food quality throughout its lifecycle, starting from purchase to disposal. It actively detects and minimizes food waste, ensuring that only high-quality and hygienic food reaches customers. Additionally, it supervises the proper disposal of expired and contaminated items in compliance with food safety standards. A real-time digital dashboard provides clear insights into each item's condition and signals timely alerts to staff when immediate action is required.

The motivations behind developing such a system are:

- Ensure food safety and public health by controlling fraudulent practices of using expired and contaminated food
- Minimize food waste, operational losses, and promote sustainable restaurant practices
- Boost transparency and enhance operational efficiency
- Establish customer trust and promote decisions based on data

The Smart Kitchen Monitoring System shows an idea designed to make kitchens safer, more efficient, and intelligently interconnected. The use of advanced technologies and creativity is driving this transformation, ensuring that innovation addresses practical as well as real-world needs. Also, the system promotes operational integrity, transparency, and the advancements help foster a safer and more sustainable food ecosystem, mitigate the environmental effects of the hospitality industry, and enhance the overall quality of kitchen inventory and dining.

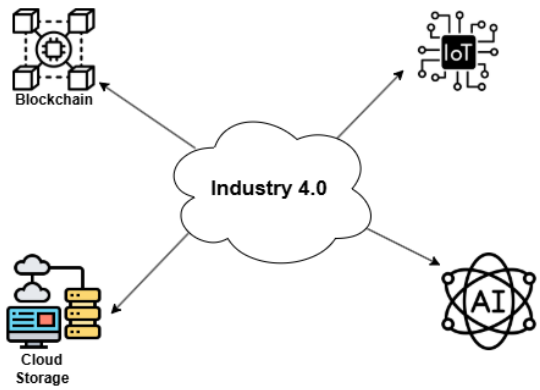

Figure 1: Smart kitchen in compliance with IOT, Blockchain, Cloud storage, and Artificial intelligence

1.1 Problem Definition

In June 2025, a DR kontant investigation and follow-up inspection by the Danish veterinary and food administration found systemic problems across all 11 restaurants in Denmark. According to several former employees from various restaurants that DR Kontant has spoken to, over the years, KFC restaurants in Denmark have used a special scam to be able to use chicken a little longer: falsified expiry-date labels on raw chicken, improper thawing/storage, and other hygiene breaches. The company temporarily closed the Danish restaurants, and the franchise relationship ended.[15]

In September 2018, a similar incident happened with the Domino's pizza on a Danish TV show called Operation X exposed the systematic re-labeling of expired food products, poor working conditions, rat infestations, and terrible hygiene at Denmark's largest pizza chain. According to their former employees, they sell meatballs and shawarma that have passed their sell-by dates.[8]

These are just two highlighted incidents because they are the biggest chain company in the world. By following, and if we search for, we will definitely be able to find more and more. As people become busier with work, they increasingly rely on restaurant and delivery apps to meet their daily nutritional needs. As a result, more restaurants are being built and more unhygienic food is being served. Although in Denmark they have established a Smiley scheme to keep monitoring the shops and restaurants. Under that scheme, they frequently visited all the stores and restaurants to give them three different smiley faces according to their food safety regulation [1]. Keeping these factors in mind, we are thinking of making

1.2 Research Questions

In order to address the challenges of food quality, wastages and using of expire food item in restaurant kitchens, this project examines the use of Industry 4.0 technologies and finally the following research questions are introduced to guide the system design and analysis of the proposed Smart Kitchen Monitoring System.

- How can Industry 4.0 technologies can be implemented to create Smart Kitchen Monitoring System that enhances food safety in restaurants?
- How can AI based image analysis and classification incorporated with sensor data like temperature, co2, gas value and humidity helps to monitor food quality and detect spoilage level?
- How Blockchain technology ensure data integrity, security and tracebility for food management in Smart Kitchen System?

1.3 Related Work

In recent times, many restaurants are concerned about the preservation of food items in their kitchen. With the rise of technologies like Internet of Things (IoT), artificial intelligence, and cloud computing, kitchens are evolving from simple cooking spaces into intelligent environments capable of tracking appliances, managing energy consumption, and enhancing food safety.

The paper [6] presents an Internet of Things (IoT)–based smart kitchen system that integrates real-time monitoring, safety control, and energy optimization. The system uses sensors for temperature, humidity, gas detection, and motion. Whole process is processed by the mobile application. Through the app, users can remotely monitor environmental conditions, control appliances, and get alerts in case of any abnormal activities. The primary focus of the work is to improve home safety, enabling remote appliance control.

In the paper [25], the study presents a smart IoT-based system that integrates sensors, actuators, and deep learning to detect and prevent food spoilage. It continuously monitors environmental factors such as temperature, humidity, and gas emissions using sensors. It shows the use of a Convolutional Neural Network (CNN) that analyzes images of fruits and vegetables to classify their type and spoilage stage. This combined data is used to predict remaining freshness time, alert users to potential spoilage, and automatically adjust storage conditions (like activating cooling or humidifiers) to maintain optimal environments.

In the paper [26], it talks about the wireless sensor system for real-time food spoilage alerts. This system detects chemical and physical changes associated with food spoilage. It is primarily focused on high-protein foods such as meat, fish, and poultry. The system targets enabling continuous, accurate, and accessible spoilage detection, improving food safety, and reducing waste by allowing early intervention.

1.4 Drawbacks and Limitations:

The primary objective of the Smart Kitchen Monitoring system is not only to prevent food from expiring but also to detect if any expired products are being used in the kitchen by restaurants. The combination of IoT, Blockchain and digital twin technology can lead to building a secure and autonomous Smart Kitchen Monitoring system. In this section, we will discuss the challenges of different related systems and how our system can solve it.

- **Data Integrity:** IOT sensors always provide continuous data to the system. This data is very important to protect from cyber threats. Storing the sensor data in the blockchain ledger would ensure the data is tamper-proof and safe to use.
- **Access Control and Authentication:** It is very important to ensure that there is specified personnel assigned for specific task to the system. Blockchain-based technology can prevent unauthorized access to IoT devices. One should be identified with the blockchain identity to ensure access to any IoT devices or the kitchen data.
- **Real Time Monitoring and simulation:** In the Smart Kitchen system, it is important to have regular update about the quality of food. Physical Monitoring cannot be effective to detect the actual freshness of the kitchen items. So digital twin technology mirrors the real time data like temperature, humidity from the IoT sensors that help to simulate the outcomes before executing the real world actions.
- **Predictive Maintenance:** It is critical to have early prediction of any upcoming defect so that it can minimize the loss and increase the lifespan of any item. Digital Twin technology would use the sensor data to detect the early signs of malfunction and alert the authorized personnel about the maintenance

1.5 Our System

Our proposed system, "Smart Kitchen Monitoring System," leverages the fundamental technologies of the Industry 4.0 revolution to introduce a fully digitalized and automated kitchen monitoring environment. The system is designed not only to detect and predict food expiration but also to generate alerts if expired or non-compliant ingredients are detected in the food preparation process. Furthermore, the system emphasizes data transparency and traceability, enhancing accountability, consumer trust, and food quality assurance within the restaurant ecosystem.

The system would integrate advanced technology aligned with the Industry 4.0 revolution. This would include technologies such as the Internet of Things (IoT), Artificial Intelligence, Cloud Computing, and Blockchain Technology. Each technology plays an interconnected role in ensuring real-time monitoring, data integrity, and food quality assurance.

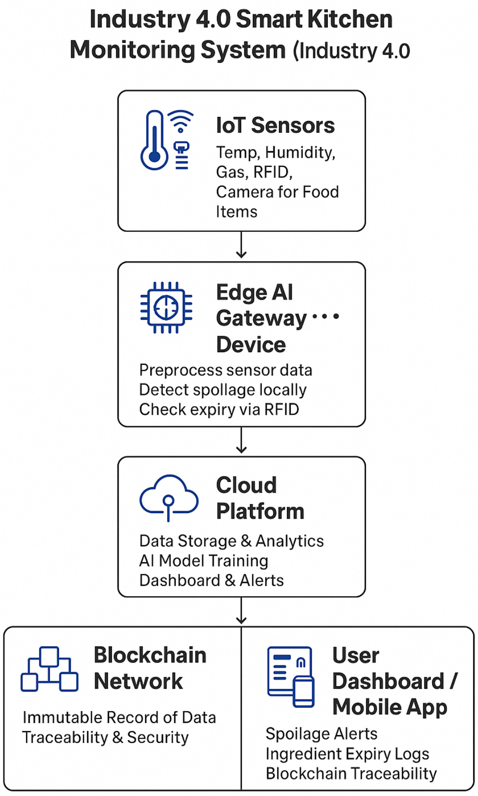

Figure 2: Workflow of different technologies[16]

- **Internet of Things :** They are used for real-time monitoring of ingredients and the environment of the kitchen. Different IoT sensors for Gas, Temperature, Humidity, and RFID tags would be attached to the storage area. These sensors would continuously transmit the data to the cloud for analysis. This sensor ensures the continuous detection of food freshness, detects improper storage conditions, and prevents the use of expired ingredients in the restaurant.

- **Artificial Intelligence:** This technology would be used for data analysis and predictive decision-making. The AI model would be trained with different Machine learning algorithms to analyze sensor data to predict spoilage and detect abnormal conditions (high temperature in the fridge and others.)

- **Cloud computing:** This technology would be used for centralized data storage and accessibility. All the IoT sensor data, AI analytics, and inventory records would be stored securely in the cloud. It also facilitates data scalability, backup, and interoperability among different restaurant branches.

- **Blockchain Technology:** This technology acts in data integrity and transparency in food traceability. All the transactions of every product would be recorded on a tamper-proof blockchain ledger. Use of smart contracts would help to verify the freshness compliance of the ingredients before using it. This technology would ensure the trust, transparency and accountability in the food supply and preparation process.

By combining this technology, we are trying to develop a cyber-physical kitchen ecosystem where the IoT provides real-world data, AI would present the intelligent insights, cloud would enable data connectivity, and blockchain would ensure data integrity and trust.

Together, this technology would create a smart, transparent, and autonomous kitchen monitoring system aligned with the Industry 4.0 goals- improving food safety, reducing waste, and enhancing customer confidence.

2 Methodology

2.1 Research Approach

This section briefly explains how we investigated the smart kitchen monitoring problem from idea to design and mock implementation. We have used design design-oriented engineering approach to solve our entire project. At first, we have analyzed the kitchen problem, then studied related work then designed and evaluated a conceptual system.

2.1.1 Main phase

- **Problem and requirement analysis:** In this phase, we have identified real food safety and waste issues. We found several exemplary food use cases and identified a lack of traceability issues, and delivered functional and non-functional requirements.
- **Conceptual and architectural design:** Then, we design the system architecture, including IOT devices, ML pipeline, cloud backend, blockchain, and dashboard. In addition to these, we created some diagrams about data flows and a sequence diagram.
- **Mock Data And AI Pipeline Exploration:** For training our model, we created a mock dataset by using the dataset on which our AI pipeline is built to test the system behavior. The entire AI pipeline includes data preparation, model selection, model training, and results.
- **Integrated concept and evaluation:** In this phase, we have explained how IOT, cloud, and blockchain can be integrated. And we evaluated the design qualitatively rather than full deployment.

This structural approach ensures that every design decision is grounded in identified pain points and tries to evaluate them against the functional and non-functional requirements.

2.2 Data Analysis

The proposed system analyzes multiple data sources that include temperature, humidity, gas sensor readings, RFID-based expiry information, as well as image data captured by cameras. However, as we are not able to collect the real-time data so we decided to create mock data for training and evaluation of the model. At first, the raw sensor data is cleaned in order to remove noise and missing values, then normalized to ensure consistency. Image data is resized and formatted to meet the input requirements of machine learning models. Also, RFID data is checked appropriately to validate product identity and expiration dates. A machine learning technique called the CNN model is applied for image processing [7] to classify food items into fresh, near-expiry, or spoiled categories. They are used to detect visual patterns of spoiled items, while sensor data helps identify unusual storage conditions. The results are analyzed, combined, and compared against predefined food safety thresholds, and when possible risks are detected, alerts are generated to support timely decision-making as well as enhance food safety management.

2.3 Ethical Consideration

Ethical considerations are the fundamental source that shapes research and ensures fairness, transparency, and respect for every party involved. While conducting research, they protect participants' rights and maintain data integrity. It plays an important role in the design, development, and implementation of the Smart Kitchen Monitoring system.

Since the project involved a huge collection and transmission of data, ethical standards were maintained throughout the project in order to ensure trust, transparency, and user privacy.

- The data that has been collected does not contain personal or confidential data from users. Privacy of the user is respected, complying with GDPR.
- The system records important information in a blockchain ledger. This prevents the unethical technique of altering information, thus promoting transparency and data integrity.
- The proposed system plays an important role in ensuring safer food consumption by minimizing the use of expired food.
- The use of technologies(IoT, AI, and blockchain) and integration are designed for safety rather than enabling surveillance or prioritizing profit.
- Minimization of food waste is also taken into consideration, which ultimately helps in reducing environmental impacts.

2.4 Project Management

This section gives the clear idea about how we take over to complete the project from the beginning to the end of the project.

2.4.1 Project Planning and Scope Definition

At the very beginning of the project, we clearly defined the scope and objectives to focus on development and the effective use of available time and resources. We first discuss looking into digital product Passport for food safety, but because of time constraints and limited resources, we limited our scope and outlined the scope of project to design a Smart kitchen monitoring system at a conceptual and architectural level, focusing on cloud-based decision making and AI-based food spoilage detection. But due to academic and practical constraints, the project aimed to test the system's feasibility through a functional prototype and detailed system design.

2.4.2 Task Breakdown and Workflow Organization

The activities of the project were divided into clear phases to maintain a structured workflow. The first stage includs problem analysis and a comprehensive literature review to gain knowledge about existing solutions and technological gaps. This was then followed by system architecture, where we conceptually define the main components of the system, including IoT integration, cloud backend, blockchain, and AI models. The second step involved a spoilage detection prototype with the use of AI. It was trained with simulated kitchen data, where mock sensor data was used to detect spoilage and image data to train the AI. Finally, the evaluation of the system, analysis of the results, and documentation of the reports were conducted to sum up the findings and describe the suggested solution in a formal way.

2.4.3 Tools, Collaboration, and Monitoring

To support effective collaboration and development, effective software tools were used throughout the project. The Overleaf platform is used for combined report writing by the team members and also continuous monitoring of report status by the supervisor. Google calendar was used for scheduling project meetings and discussions wth the supervisor. Python and PyTorch were employed for implementing and testing the AI pipeline, while data handling and preprocessing were managed using standard data analysis libraries (like pandas and NumPy).

2.4.4 Risk Management and Constraints Handling

Some limitations and threats were identified during the project, such as a lack of access to the actual IoT sensor data, absence of physical kitchen hardware, and time constraints during the academic semester. In order to reduce these issues, synthetic and mock datasets were created to simulate sensor measurements and food spoiling scenarios, allowing controlled experiments and verification of the AI pipeline. Cloud, blockchain, and digital twin components were conceptually modeled, and the logic of the system and the flow of data were clearly defined, even though they were not actually implemented. These risk mitigation measures enabled the project to meet its goals and still preserve technical validity and academic rigor.

3 Theory and Background

This section highlights the required theoretical ideas and concepts that form the base for the Smart Kitchen Monitoring System. It shows the function of modern technological innovations like the digital Things Technology, Internet of Things (IOT), blockchain technology, Cloud Storage and computing, and data management, powered by Industry 4.0 principles that help in enhancing food safety, inventory management, as well as operational efficiency within restaurant kitchens.

Industry 4.0

The term Industry 4.0 was first used in 2011 as part of a German government high-tech strategy program, but it only gained popularity after the World Economic Forum adopted it in 2016 [30]. Also known as the Fourth Industrial Revolution, is characterized by the incorporation of advanced digital technologies into manufacturing and industrial processes through the collaboration of smart technologies such as cloud computing, AI, analytics, IoT, and digital twin[14].

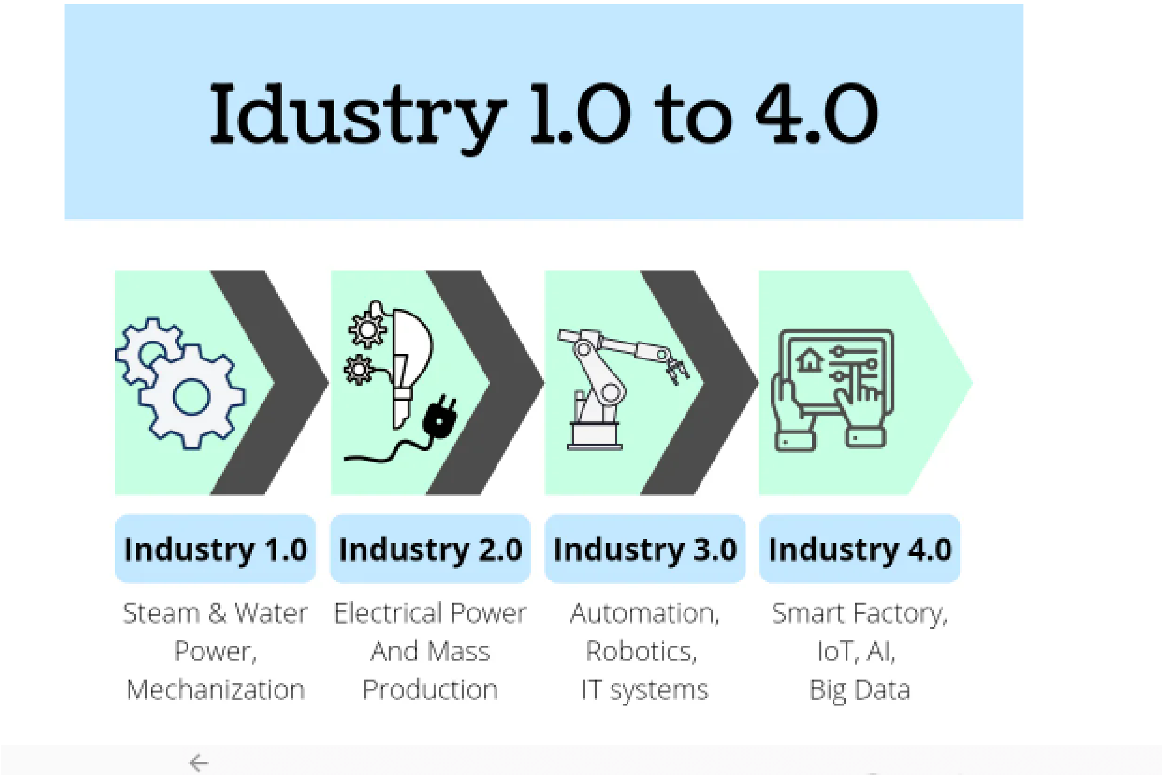

Figure 3: Industry 1.0 to 4.0 [13].

Similar to the preceding industrial revolutions that transformed work methods through the use of machines, electricity, and computers, Industry 4.0 marks the next significant evolution in industry. Below described are the industrial revolutions that describe major periods of technological and social transformation in human history and are driven by innovations.[14]

- First Industrial Revolution: In the early 1800s, the First Industrial Revolution started, and the steam engine reduced the dependence on human and animal work in industry, marking the beginning of a new era in manufacturing and industry processes.
- Second Industrial Revolution: A century later, the second industrial revolution was introduced, and the use of petroleum products, gas, and electric power was increasing. With the help of these new power sources and communication mechanisms through telephone and telegraph, manufacturing processes saw the advent of mass production and a certain level of automation.
- Third Industrial Revolution: It started in the mid-20th century and is also called the digital revolution, incorporating computers, advanced telecommunications, and data analysis into manufacturing and industrial processes. In this timeframe, relatively large computers that were simple in design were created, which provided a solid basis for the evolution of modern-day machines. [13].
- Fourth Industrial Revolution: Industry 4.0 represents the industrial revolution that is currently being put into practice in today's world, characterized by smart machines that can trigger action and controls. Efficiency has been improved so that it can meet customer expectations. Everything is completely digitalized in the manufacturing process, and more advancements are expected to be made. [13]

3.1 Digital Twin Technology

Numerous creative concepts and cutting-edge technologies have emerged as a result of the worldwide digital transformation pushed by Industry 4.0, with Digital Twin Technology drawing considerable interest and now expanding across various domains, including healthcare, logistics, and smart environments. Powered by the fundamental principles of Industry 4.0, digital twins have developed into a crucial facilitator of smart systems [22].

A digital twin is a virtual representation of a physical object or system that reflects its complete lifecycle and is continuously updated with real-time information. It uses simulation, machine learning, and data-driven reasoning to aid in informed decision-making and performance optimization [5]. It means creating the virtual model that is a similar (twin) of the physical object. The object could be a vehicle, a building, or an aeroplane engine. Sensors that are connected to the physical object gather data that can be reflected in the

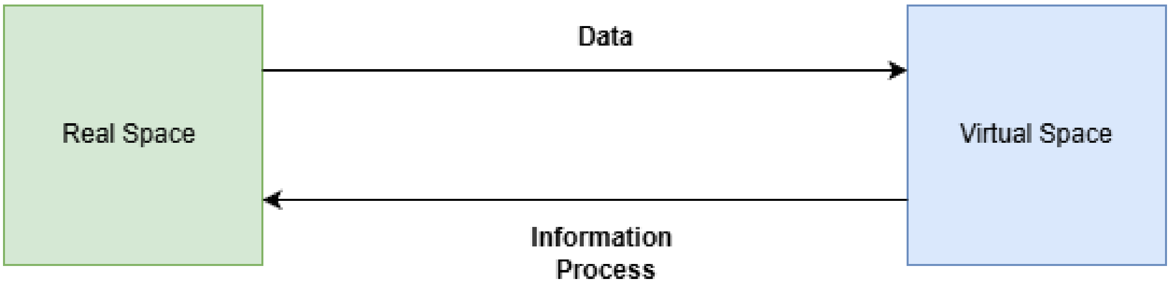

Figure 4: Conceptual Model For Digital Twin [27].

virtual model. So, the digital twin provides anyone who views it with crucial data regarding the performance of the actual object in the real world. With this technology, users and stakeholders can gain valuable insights into a system's core internal processes, the interactions between its components, and the expected behavior of its physical counterpart [4].

Figure 1 (Conceptual Model of Digital Twin), introduced by Professor Michael in 2011, gives the early model of the Digital Twin concept that demonstrates how physical and virtual representations of a system interact through data and information flow. It is not just a model but an integrated system that continuously links the physical (real space) and digital (virtual space) through data exchange by ensuring a correct, real-time representation and effective optimization of the physical system.

3.2 Internet of Things(IOT)

Internet of Things gave us the freedom to connect and share data with other devices and systems over the internet through sensors, software, and other technology. In our proposed smart kitchen monitoring system, we plan to use some advanced sensors to detect and monitor the food life cycle. One of the most advanced sensors for collecting more information about food safety and quality in real-time is the RFID sensor. In this section, we are going to discuss the RFID and how it works, and how it will be a suitable match for our proposed system.

3.2.1 Radio frequency Identifications(RFID)

RFID is a non-contact technology that can recognize a specific target by using radio frequency. It has three main components RFID tag, a reader, and an antenna with an application system. An RFID tag is a chip with a unique identification number, which can be read and stored through the RFID reader, and the antenna is used to transmit radio frequency between the tag and the reader. Currently, it is widely used for food packaging and storing in the warehouse [31]. In Figure 5 shown how an RFID sensor can be placed in a warehouse door to monitor hundreds of food packages at a time. As we are thinking of establishing a system to monitor the kitchen food cycle, we are aiming to use an RFID sensor to track our food life cycle inside the kitchen, freezer, and waste management area to ensure proper food hygiene.

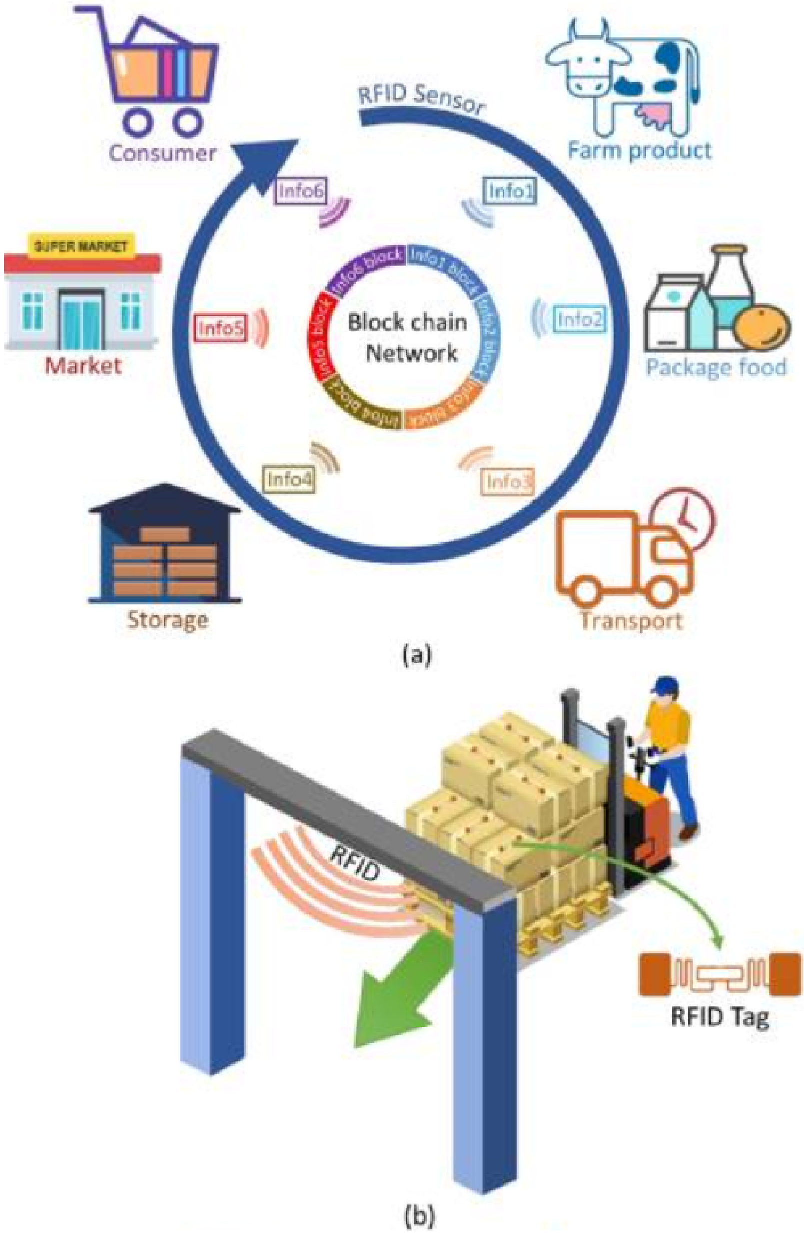

Figure 5: Architecture of food supply chain based on RFID and the Blockchain technology[31]

3.2.2 Temperature Sensor.

Temperature control is one of the most critical factors in food safety and hygiene. The temperature sensor provides real-time monitoring to ensure that food is stored and processed at a safe temperature, preventing microbial growth. According to the World Health Organization, a temperature below 5°C will protect all the food, but it will spoil gradually. The temperature between 5°C to 60°C is considered to be the most dangerous zone for any food[23]. We aim to utilize an advanced temperature sensor to monitor food temperature and transmit the data to our cloud storage for further analysis.

3.2.3 Humidity Sensors

Humidity control is another critical factor in food safety. If the food is not stored at an ideal humidity level, it can lead to moisture loss to wilting, and drying out of food and vegetables. According to the International Journal for Food Microbiology, bacteria grow when relative humidity exceeds seventeen percent, and mold growth can begin at levels above sixteen percent [12]. So it is really essential to control the humidity to create a safe food environment, reduce contamination risks, and increase shelf life. So we are planning to use humidity sensors to enable real-time monitoring, predictive maintenance, and data-driven alerts, making them a vital component along with the temperature sensors.

3.2.4 Air Quality, CO2 Sensors.

In a smart kitchen area, air quality is not just about the comfort-it is directly involved with the food safety, staff health, and regulatory compliance. As we have seen in Denmark, most of the kitchens use a CO2 sensor to detect the CO2 level inside their freezers to keep monitoring. Gases like CO2 and VOCs cause microbial contamination, reduce indoor air hygiene, and even alter food quality. According to the World Health Organization, poor indoor air quality can lead to serious health issues, and it's a major risk for contamination in food environments [10].

3.3 Blockchain Technology

Blockchain technology has emerged as a backbone of the Industry 4.0 revolution, providing a secure, decentralized, and transparent mechanism for recording and managing digital transactions. Blockchain stores data across a network of interconnected nodes in a distributed ledger. It ensures that every record is immutable and verifiable and most importantly, blockchain's decentralized architecture ensures data integrity, traceability, and trust among all participants within the system. [17]

Blockchain is the base of our system, assuring food safety, transparency of quality, and regulatory compliance. With metadata covering the supplier's details, the date of manufacture, the storage conditions, and the expiration date, every ingredient or food item would be recorded on the blockchain. Every transaction or change in these items as they flow through the restaurant's supply chain, from buying to storage and use, is safely documented on the blockchain ledger. This would ensure that all food-related data is permanently recorded and transparent, making it nearly impossible to alter or fabricate expiration dates. [20]

Overall, the use of blockchain technology in the Smart Kitchen Monitoring System establishes a trustworthy digital infrastructure that ensures the principles of transparency, accountability, and automation. By ensuring food quality and authenticity, it not only increases customer confidence but also supports Industry 4.0's broader goals of a more effective, legal, and sustainable restaurant business.

3.4 Cloud Storage and Computing

Cloud Computing refers to he provision of computing services such as storage, processing power and data management over the internet on an on-demand basis. Cloud platforms offer flexible and scalable resources that are accessible at any time or place, rather than relying on local servers and physical infrastructure. Cloud storage offers centralized and secure data storage, which allows storing large amounts of data, retrieving and also efficient management.[19]

Cloud computing is essential in the context of Industry 4.0. It can be applied to execute real-time data processing, facilitate the interoperability of the systems, and support intelligent decision-making. Smart environments generate a continuous flow of data from IoT devices, sensors, and machines, which demand scalability and robust computing capacity. Cloud platforms offer the required infrastructure to manage this data, use advanced analytics, and combine artificial intelligence models without the constraints of local hardware. [14]

Moreover, cloud computing serves as an integration layer between the physical systems and digital services. It allows the interaction of IoT networks, AI-driven analytics and other digital technologies, including blockchain and digital twins, to interact seamlessly. Cloud computing provides the backbone of contemporary smart systems by providing high availability, fault tolerance, and international accessibility to aid data-driven, autonomous, and efficient operations. [22]

3.5 Data Management

Data management plays an important role in modern smart systems. It involves the secure collection, processing, and utilization of data to enhance business outcomes. An effective data management technique includes integration and storage of data across multi-cloud environments that ensures availability, security, and leverages AI-driven automation for data analysis and decision making. [29].

Effective data management is important for the successful operation of cyber-physical convergence, digital twins, and the required objectives of Industry 4.0. Data serves as the cornerstone connecting the physical and digital realms, facilitating seamless interaction among machines, systems, and people. Within the framework of I4.0, appropriate data gathering, storage, integration, and analysis guarantee that digital models precisely depict real-world situations and that decisions made in the cyber domain can be relied upon to enhance physical operations. Therefore, it is said that in the absence of robust data management, information can turn incorrect, which hinders the ongoing feedback loop essential for a smart system.[24]

3.5.1 Role of AI in Data Management

In the current era of big data, the volume of information generated in scientific research is increasing at a faster rate. It can be quite challenging to extract valuable insights from this data. The expansion of data creates new insights and knowledge, but also generates significant challenges. Traditional data analysis methods are unable to cope with the complexity and volume of modern datasets. To overcome these challenges, the use of Artificial Intelligence has become an important tool in the scientific community.[]

A vital function of Artificial Intelligence is the improvement of data management. Organizations can leverage AI and ML tools to utilize their data effectively by optimizing tasks like data source integration, data cleaning, and data retrieval. This enables companies to make data-driven decisions.[28]. AI encompasses the wider domain of creating systems that can mimic human intelligence, while machine learning, a subset of AI, is dedicated to enabling systems to learn from data so, AI can analyze large datasets, recognize patterns, and forecast potential risks. [11]. Also, one of the functions of AI is to convert raw data into valuable real-time insights that help in making informed decisions based on data, ensuring growth, innovation, and efficiency [18]. The integration of AI with data management enhances accuracy, facilitating smarter, more self-directed operations that align with the goals and objectives of Industry 4.0.

4 Analysis

4.1 Problem Definition and Objectives

In this modern time, the food industry faces various challenges related to food safety, ingredient traceability, and waste management. In various commercial kitchens, manual monitoring of ingredient freshness and expiry might result in inefficiencies and health risks. The proposed system introduces Industry 4.0 technologies. The collaboration of different technologies like the Internet of Things (IoT), Artificial Intelligence (AI), Cloud Computing, and Blockchain could lead to the development of an integrated Smart Kitchen Monitoring System. The system's primary objectives are to:

- Detect the spoilage of food in real-time using sensor and image data.
- Identify and prevent the use of expired ingredients during food preparation.
- Provide secure and transparent data storage using blockchain technology for auditing and compliance.
- Improve operational efficiency and food safety through intelligent decision-making.

4.2 Functional and Non-Functional Requirements:

Functional and Non-functional requirements define a system's general functioning and quality. Functional requirements define how the system should respond to inputs and manage various operations. They emphasize specific features and processes that the system must carry out. On the other hand, non-functional requirements cover factors such as performance, security, dependability, and usability, defining standards for how successfully these functions are carried out. Non-functional requirements ensure the system operates effectively, safely, and with a positive user experience, whereas functional requirements guarantee that the system produces the desired results. Both are necessary to create a reliable, user-friendly, and functional system.

Some of the functional and non-functional requirements of the proposed system are listed below:

Functional Requirements:

- The system should continuously monitor the conditions of food and ingredients using IoT sensors.
- The system should detect food spoilage through AI-based analysis of temperature, gas emissions, and visual data.
- The system should validate ingredient expiry dates using RFID/NFC tags.

- The system should generate real-time alerts to kitchen staff and supervisors when anomalies or violations occur.
- The system should securely record data on food handling and ingredient usage within the blockchain network.
- The system should perform visualization, analytics, and reporting of results in a Cloud-based dashboard in the restaurant.

Non-Functional Requirements:

- The system should effectively handle multiple kitchen nodes and sensor inputs.
- The system should have minimal downtime for data collection and communication.
- The system should encrypt all transactions and data transfers.
- The dashboard should be user-friendly for non-technical kitchen personnel.

4.3 Service Architecture.

In this section, we will present and define the proposed system architecture, including its appearances as a combined system. Figure 6 displays our complete system architecture.

In our system architecture(figure 6), on the left side, we have our Data source devices, such as temperature, humidity, and CO2 sensors, alongside RFID readers to track conditions and inventory in real-time. At the same time, storage units such as fridges, freezers, and shelves serve as sensor-equipped repositories that continuously report the status and environment.

The green box in the middle, between Data sources and the cloud backbone, is the Edge gateway. It will function as a Data ingestion layer between those two layers. It will receive data from sensors, preprocess it by cleaning, filtering, and formatting, and transfer it to the main backbone for deeper analysis.

At the center of the figure 6, the big blue box is your main backbone. It will function as the system's intelligent layer by aggregating heterogeneous data streams from connected devices, executing AI and machine learning pipelines for food recognition, generating shelf-life predictions based on temporal and environmental features, and performing continuous, real-time monitoring of operational states. This platform is also responsible for managing both structured and unstructured information, enforcing service logic to trigger alerts, automatic actions, and supporting the decision-making process.

Right next to the cloud backbone, we have our blockchain layer, which ensures trust, transparency, and security across the entire food monitoring process. It ensures that sensitive data, such as expiry dates, sensor logs, spoiled alert, and food movement records, remain tamper-proof, so that no one can delete or modify them. Every action will be recorded with a timestamp to build transparency between multiple parties inside the kitchen to see exactly:

- When food was stored.
- What conditions does it expire?
- Who handled it.

On the right side, we have our display and the smartphone, which receives notifications and responds to them. The display will be used to interact with customers and showcase alerts about food hazards. Additionally, a smartphone can be used during the kitchen closure period to stay connected to the system.

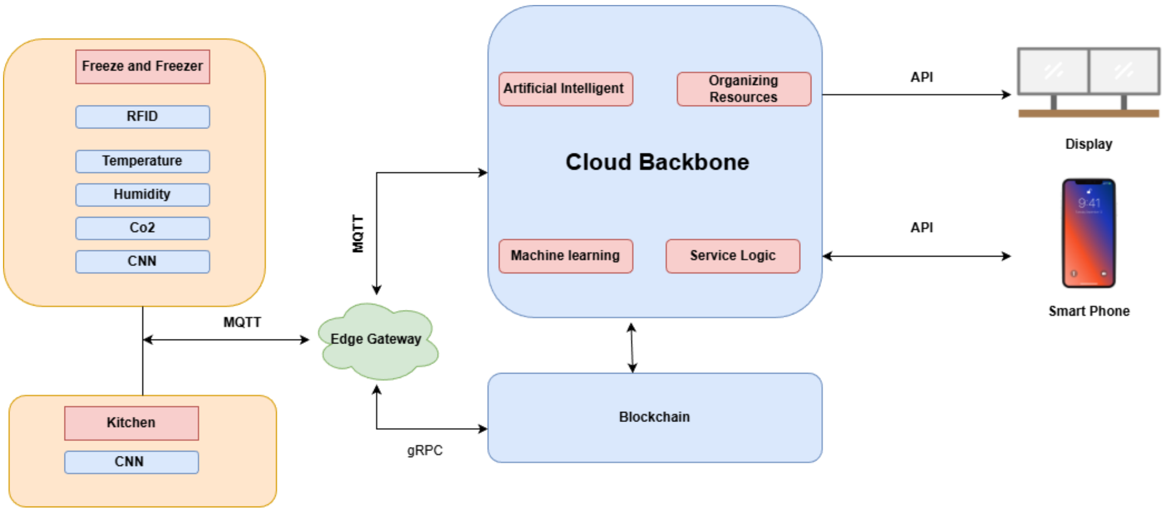

Figure 6: Service Architecture.

4.4 IoT Device and Data Flow Analysis

In our proposed system, we are planning to use IOT sensors to collect real-time data from the different storage locations. After collecting the data, they will be compared with the standard value to create a hazard alarm for the kitchen environment. Figure 7 shows how data from the kitchen storage will be collected through IOT sensors. The IoT devices capture heterogeneous data types such as numerical(temperature, humidity, and CO2), categorical (RFID IDs), and visual (images). These data are transmitted to the edge gateway using the MQTT protocol for lightweight, reliable communication. Edge-level AI pre-processing filters out redundant or stable readings, minimizing cloud upload volume. This architecture ensures low latency, energy efficiency, and real-time decision-making, which are essential in kitchen environments where rapid spoilage detection is critical. All the collected data from the sensors will be stored in the cloud storage for further processing.

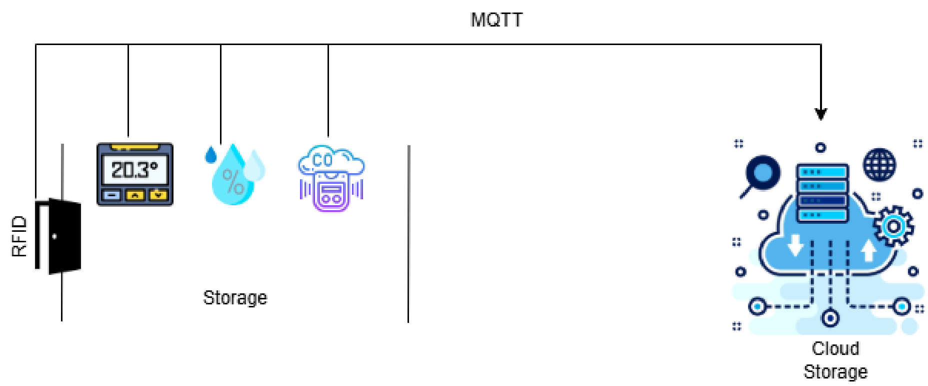

Figure 7: IOT Sensors Data Flow Diagram.

4.5 AI Spoilage Detection Model Analysis

AI provides effective approaches and acts as a decision making layer to analyze and enhance food spoilage detection methods using extensive datasets and sophisticated algorithms. The technologies and algorithms can analyze complex patterns and correlations from data collected from sensors to identify signs of spoilage more efficiently and in an earlier phase than traditional methods.[3]

The AI model for the Kitchen Monitoring System mostly uses supervised learning models such as Random Forest (RF), SVM, and Artificial Neural Networks (ANN), as it is trained on a labelled dataset. The label is distinguished as 'fresh', 'spoiled', or 'going to decay(expire)'. These labelled data are based on images, temperature, humidity, gas values, and microbiological conditions. During training, the system learns the pattern and can predict spoilage status for unseen and new data.

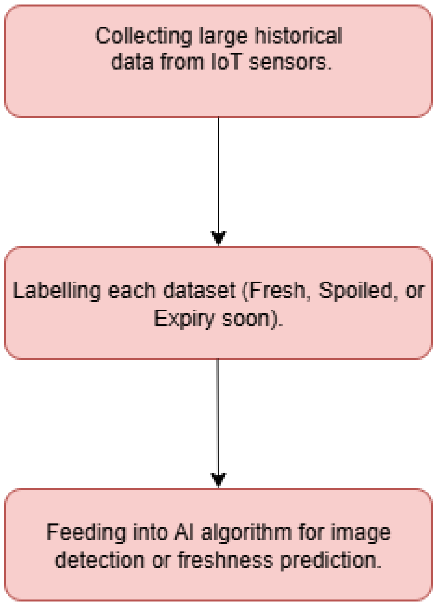

Figure 8: Training Process

The above figure shows the training process involved. It starts by collecting large amounts of historical data from IoT sensors and cameras. After that, it labels each dataset entry with the correct spoilage state and feeds it into AI algorithms such as convolutional Neural Networks (CNNs) for image-based detection[21] or Random Forest or Support Vector Machine (SVM) for diverse food products and freshness prediction[21]. Finally, if the spoilage is forecasted, an alert is sent on the cloud dashboard, and the system can record the result in the blockchain ledger for traceability.

The AI model periodically re-trains itself using newly collected data that is stored on the cloud, and the process allows the model to adapt to varying environmental conditions, storage habits, and different types of food. After periodic training and adjustment to diverse conditions, the system improves its performance and maintains high accuracy over time.

4.6 CNN Camera Integration for Visual Detection

A Convolutional Neural Network (CNN) camera is integrated in both refrigerators (Freezer) and the kitchen. It plays an important role in visual food inspection and continuously captures images of food items. With the help of deep learning, it classifies the condition of items such as a change in color, mold formation, etc The images that are captured are processed on the edge platform and perform basic pre-processing steps, and then sent to the cloud, where the image is classified into different categories such as fresh, spoiled, or going to expire. The results are integrated with the sensor data, such as temperature, humidity, or gas values. for instance, if both humidity and CNN detect the sign of spoilage, the system triggers an alert and warning to the cloud dashboard.

In order to ensure traceability and immutability, each and every classification outcome and alert is securely logged in the blockchain ledger. This ensures that the visual presentation of the image cannot be modified by staff or a person without authorization. Overall, the intelligence of the system is enhanced by the CNN camera, which provides visual validation that goes beyond the capabilities of IoT sensors.

4.7 Cloud Storage and Computing Framework

The Cloud Backbone, as shown in (Figure 6), is the intelligence and coordination center of the proposed smart kitchen monitoring system. It obtains real-time data of environmental and objects of distributed kitchen sensors using an edge gateway based on the MQTT communication protocol. The integration of Artificial Intelligence (AI) and Machine Learning (ML) modules processes the multi-sensor inputs to predict food spoilage, detect expired ingredients, and regularly monitor kitchen conditions. Deep learning models, particularly for Convolutional Neural Networks, improve the accuracy of spoiling detection by identifying patterns in sensory data and camera inputs. Service Logic and Resource Orchestration are also found on the cloud and guarantee computational scalability, load balancing, and continuous operation with dynamically varying workloads. Processed data and results of inference are stored in the data format of safe clouds, and the API layer transfers results to user-interfaces, such as smart displays and smartphones, to realize real-time visualization, notifications, and decision support. The event-driven nature of the cloud makes sure that in the case of a deviation of the environment, the corresponding workflow (alerts dissemination or automated maintenance routines) is triggered in time.

4.8 Blockchain Integration Analysis:

The system is based on the Blockchain Layer to provide trust, transparency, and integrity of data through keeping an unchangeable record of significant kitchen and freezer processes. The blockchain is able to store hashes of cryptographic data and metadata of data processed in the cloud, rather than storing raw high-volume sensor data, which allows for verifiable proof of authenticity with maintained efficiency. Such distributed registry ecosystems remove the points of failure and help in tamper resistance because every transaction or update is authenticated on many nodes before being committed.

Smart Contracts are self-executing programmes that are used to specify business logic. They are used in the blockchain to carry out automated verifications of operations, including checking ingredient expiration, verifying temperature deviations, or authenticating devices. At the IoT component and edge gateway level, the IoT components and edge gateways receive a Decentralized Identifier (DID), and through blockchain-based credentials, they can verify each other and control access to cloud services based on these credentials, rather than centralized authorities. The cloud and blockchain communicate through gRPC or a Web3 interface, as this is a secure and low-latency method for data exchange. These mechanisms mean that blockchain technology offers the background of decentralized jurisdiction, traceability, and auditability of every food-handling and environmental information in the kitchen ecosystem.

4.9 Integration of Cloud and Blockchain in Smart Kitchen Monitoring System

Integration of Cloud and Blockchain provides a unified system that combines the computing savvy of the cloud computing with the decentralization of trust and immutability of blockchain technology. Upon sensor data being run through the AI and ML models of the cloud, the resulting information, say spoilage prediction or expiration confirmation, is hashed and anchored to the blockchain to be permanently verified. Blockchain transactions are initiated by the cloud through smart contract calls and allow conditional automation, using established data states. For instance, when the blockchain records an ingredient with an expiration date that has passed or when environmental requirements have been violated, a smart contract may emit an event such as "Food Expired" or "Spoilage Confirmed." Those events on the blockchain are subscribed to by the cloud backbone that then initiates appropriate service logic to create user alerts, refresh dashboards, or modify the operational parameters. This is a two-way communication to make sure that every decision that cloud based intelligence has made can be cryptographically verified and has integrity restrictions that are enforced by blockchain. Under this unified design, high speed computation, scalability, and real time analytics are provided by the cloud, and trust, accountability, and decentralized data validation are guaranteed by the blockchain. They are combined to create a stable, intelligent, and transparent digital infrastructure of continuous food safety tracking, automated management, and reliable decision making of a smart kitchen ecosystem in the future.

4.10 Data Security and Privacy Considerations

Ensuring data security and privacy is important for the kitchen monitoring system, as this system incorporates various technologies such as IoT, Blockchain, Cloud storage, and AI. The continuous stream of data from sensors, information about ingredients, and records should be protected and tamper-proof, as there can be introduction of vulnerabilities and weaknesses such as unauthorized access, manipulation of devices, Man-in-the-Middle (MITM) attacks, data privacy leakage, and corrupted data that reduces AI model accuracy. To avoid misuse, robust security measures must be implemented throughout all technological layers of the system, which are as follows:

- All the communication processes between IoT devices are encrypted well and device authentication is employed in order to decrease unauthorized access.

- The implementation of blockchain to ensure that records of temperature logs, expiry checks, and spoilage alerts cannot be altered.
- Performing periodic auditing in cloud storage
- Ensuring compliance with GDPR by preventing the collection of personal data, setting strict obligations for storing or processing it and maintaining privacy safe AI training practices.

5 Design

5.1 Data Flow Diagram

This section presents the system architecture of the proposed AI-driven Food Spoilage Detection and Traceability Platform. The design combines different data feeds, graph intelligence,spoilage level, decision engines, alert service front-ends, and blockchain-supported audit reporting. The architecture aims to provide a precise, secure, and real-time system for detecting, authenticating, and reporting spoilage cases in food storage and handling facilities.

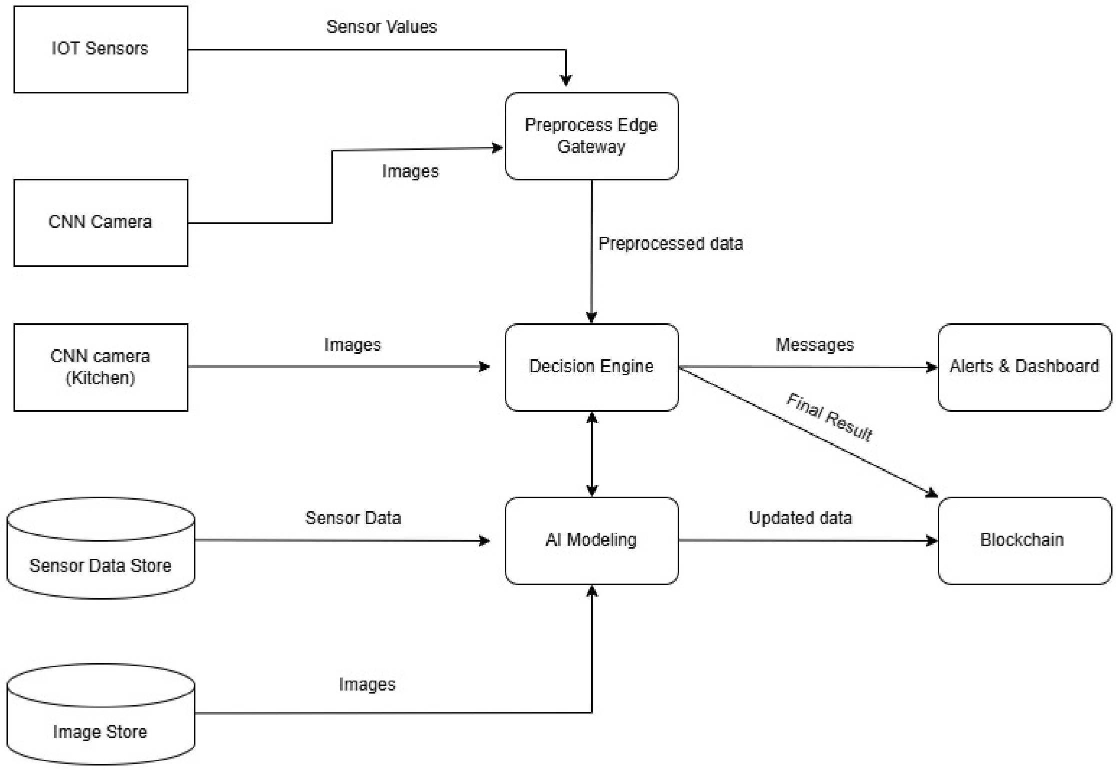

Figure 9: Data Flow Diagram

The general architecture consists of 5 main layers: data acquisition layer, edge preprocessing layer, intelligence and validation layer, decision and notification Layer, and the blockchain traceability layer.

Data Acquisition Layer has the role of capturing multimodal inputs of the IoT environmental sensors as well as the camera imaging devices. An IoT network of sensors checks vital parameters of storage like temperature, relative humidity, gas concentrations (e.g. CO2, VOCs, ethylene), pH, and light intensity. These sensors are periodically used to pass structured numerical measurements to the upstream gateway over lightweight communication protocols(like MQTT).

All raw sensor and image data are transferred to the Edge preprocessing gateway in order to trim network bandwidth consumption and deliver low-latency analysis facilities. This gateway does data cleaning, normalization, synchronization, and feature extraction.

The Decision Engine serves as the central reasoning unit that fuses outputs from the sensor subsystem, expiry validation results, and AI-based spoilage predictions. It adopts a hybrid decision model that consists of rule-based logic, threshold-based, and temporal anomaly detection, and probabilistic confidence scoring. This helps the engine to realize that there is a spoilage event that needs an urgent response, that there is a risk of future issues due to deviations in environmental parameters, and that a blockchain logging event is needed.

Both historical sensor logs and image data stored in respective repositories (Sensor Data Store and Image Store) would be referenced to detect recurring patterns or contextual dependencies.

The alert subsystem is composed of the notifications of the expiry validation process, the AI spoilage classifier, and the Decision Engine. It sends real-time messages through various means like push notifications, SMS, email or even visual messages on the dashboard screen. The dashboard offers a single user interface that will show trends of sensors, code identifications of spoiled goods, expiry alerts, annotated images, and logs of past events. The objective of this subsystem is to provide both human intervention and operational visibility at the right time.

The architecture incorporates a Blockchain Module to ensure tamper-resistant auditability and transparency of traceability. Each time one of the key events is signaled, like identification of food spoilage, environment boundary breach of the environment threshold, or product expiry, the Decision Engine starts a blockchain transaction consisting of a hashed summary of the event, time, sensor readings, and product identities. The blockchain ensures:

- Data immutability for compliance and investigations
- Non-repudiation of logged spoilage incidents
- Traceable provenance across the food supply chain
- Cryptographically verifiable audit trails

The system is made to work as a coordinated pipeline where raw sensor and image data are collected, then after edge processing is done, and finally assessed with expiry models and AI classifiers. The Decision Engine then sums up all the analytical outputs and decides what to do, such as alerting and blockchain logging. The architecture is a multi-layer solution, offering an all-around basis of food spoilage detection, awareness, and traceable documentation in real time.

5.2 Sequence Diagram

This section shows how data will flow through our smart kitchen monitoring system. It basically tells the runtime story of how the system will detect spoiled food and warn staff step by step [10].

First, the LOT devices continuously transmit temperature, humidity, and CO2 parameters into the system, which reflect the real-time conditions of every food item. Meanwhile, the CNN camera will capture the real-time images of the food, and the RFID sensor will check the tag on the food packages for any expired items.

All the input data is entered into the cloud server for pre-processing, where it is filtered out of the noise, normalized sensor values, and cropped and resized images to fit the input format that is required by the machine learning model. Then the cloud transfers the pre-processed data to the machine learning algorithm.

The fused data is input into the ML algorithm to predict the probability of being spoiled for each item. Typically, it produces a provability score or a class label such as fresh, warning, or spoiled. Then the ML algorithm will send this result to the decision engine to compare it with the blockchain-pre-stored standard spoiled threshold, generating an alarm.

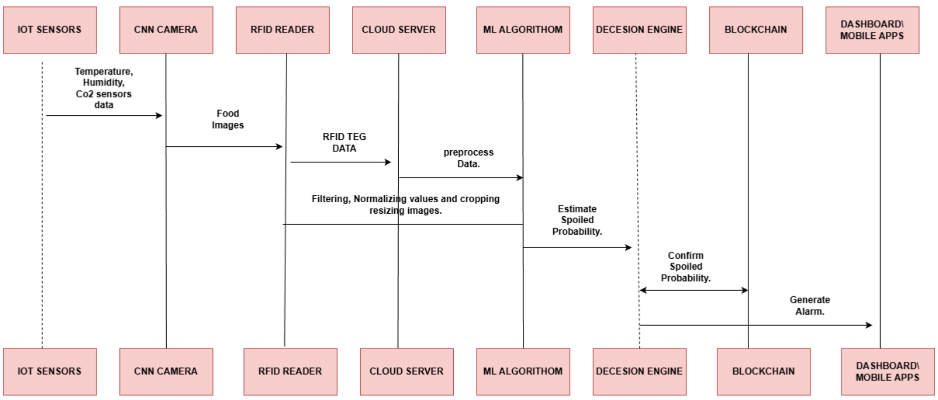

Figure 10: Sequence Diagram.

5.3 Cloud Backend Conceptual Design

This section represents the logical structure and internal flow of cloud layer of the proposed Smart Kitchen Monitoring System. This sections aims to describe how the preprocessed data obtained by edge gateway is analyzed, validated and convert to the actual decision in the cloud. This section main focus on conceptual stages of the process that allow to detect food spoilage in real-time, generate alerts and records the event securely.

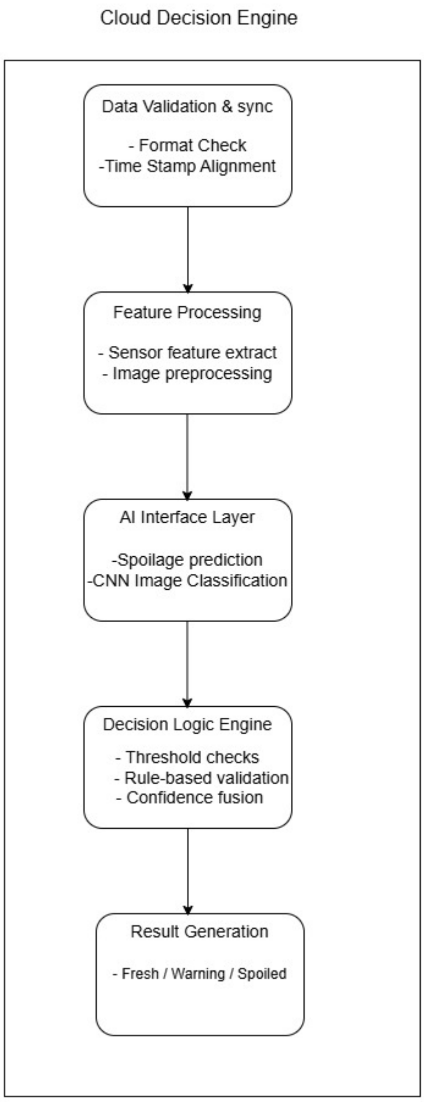

Figure 11: Cloud backend data flow diagram

According to (figure 11), once processed data are received in the cloud-based Decision Engine, they are first validated and synchronized to check consistency in format and timing of the incoming data from multiple sources. Then, all relevant data are extracted from sensor values and camera input to prepare them for analysis. Then, in the AI interface layer, ML models and CNN based classifiers would evaluate food freshness and spoilage conditions. The resulting predictions are then sent to the decision logic module, where they are evaluated with predefined thresholds and rule-based validation to determine the final system outcome. Based on the decision, the cloud would generates final results, including real-time alerts in the dashboard or kitchen staffs and cryptographically verifies events to blockchain layer for secure and immutable record keeping. This hierarchical cloud architecture provides proper decision making, scalability, and visibility in the smart kitchen ecosystem.

5.4 Prototype Scope and Implementation Boundary

The proposed Smart Kitchen Monitoring System is a comprehensive Industry 4.0 solution that combines various technical elements, including digital twin, cloud computing, blockchain technology, IoT sensors, and artificial intelligence. However, because of time constraints, infrastructure limitations, and the academic focus of this project, only a functional prototype has been implemented to validate the core intelligence of the system not the full physical deployment.

The developed prototype primarily focuses on AI-based food spoilage detection, which is considered as the central decision-making unit of the system. In place of actual IoT sensor feeds and actual camera hardware, a set of mock sensor data and image datasets is created to create simulated kitchen conditions and food decay situations. This model enables a scientifically determined experimentation and demonstration of the AI pipeline without conflict with the suggested system architecture.

Other components like IoT sensors, edge gateways, cloud services, blockchain, and smart contracts are developed at an architectural and conceptual level, represented through diagrams and workflow descriptions. All their functional behavior is modeled through an AI pipeline and result recording, but not physically executed in this prototype.

This scoped implementation ensures the accuracy of food spoilage classification, system feasibility, and decision logic without deploying any hardware or live integration into the cloud-blockchain. The full scale implementation of real-time sensor networks, decentralized blockchain infrastructure, and digital twins technology are termed as a future work beyond the scope of this project.

6 Implementation

This section describes how the Smart Kitchen Monitoring system prototype is developed and implemented, as explained in section 4 (4.5 and 4.6). The main goal is to create a mock following an AI pipeline and train a CNN model to classify food item as fresh, spoiled, or warning.

6.1 Mock Data and AI Pipeline

This section focuses on creating our mock dataset. We created the mock dataset, preprocessed, trained, and tested it in our model.

6.1.1 Creation of mockdata

To develop a smart kitchen monitoring system, a mock dataset is designed, and an AI pipeline is built to test the system's behaviour. To generate mock data, about six hundred synthetic food samples are created, and they are classified as "fresh", "warning", and "spoiled", with 200 each. Each sample contains food images(using Python and the PIL library) of 128*128 that show spoilage level, which are Fresh(green, very few spots), warning(yellow, moderate spots), and spoiled(dark spots with brown background). The goal is that the model can learn what spoilage may look like in the real world.

For each image, the row of mock sensor values, as explained in section **??**, like temperature, humidity, CO2 level, item type, storage location, hours since delivery, and expiry time, is designed. These values have a range depending on the factors such as :

- Fresh Item: low temperature and low CO2.
- Warning Item: going to expire, a bit higher temperature, and CO2.
- spoiled item: negative expiry hour remaining, higher temperature, and 2 with higher mold score.

So, this helps to show the real-world behaviour and keeps data consistent.

6.1.2 AI Pipelining

After the creation of mock data, an AI classification pipeline using PyTorch was built. The main aim of this model is to distinguish each item into one of three categories of fresh, warning, or spoiled. The following steps were carried out:

Data Preparation: The dataset is shuffled, and train train-validation split of 80 percent and 20 percent is done. While doing that, the images are resized, transformed into a tensor as well, and normalized. In order to load images and labels, a custom dataset class in PyTorch is implemented.

Model Selected: After the data has been prepared, a Convolutional Neural Network (CNN), as explained in 4, is implemented because it is mostly used for image classification. The model continues to learn and detect the visual behaviour like color formation, mold, spots, etc, associated with each spoilage level.

Model Training and Results: The model is trained using cross-entropy loss, Adam optimizer with five training epochs. Because the synthetic images are simple, clear, and can be distinctly separable, the model achieved an accuracy of 100 percent on both training and validation sets, which shows the pipelining is working smoothly without error.

After training and validation, the AI pipeline is tested to check if it's working correctly and end-to-end. For this random image has been selected with their path with their true label and predicted label as Fresh. The following output is shown:

- Fresh: 1
- Warning: 0
- Spoiled: 0

This output reveals the model is identifying randomly selected images, such as fresh with very high confidence.

6.2 Main tools and Libraries

The implementation was done in Python. Main tools and libraries are Python 3.9, PyTorch, torchdivision, Pillow(PIL) for synthetic image, and pandas for data handling.

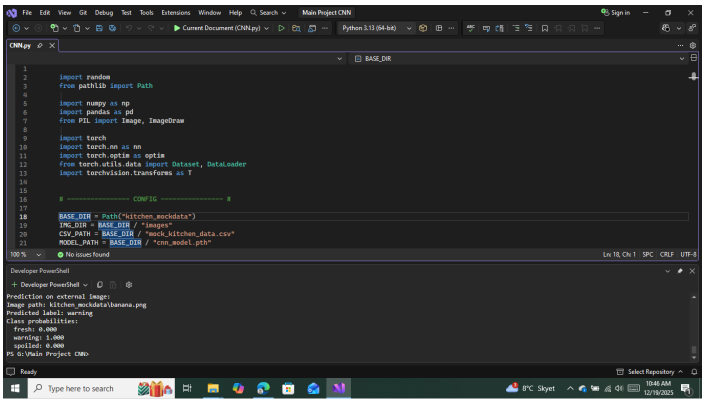

Figure 12: Tools and Libraries.

7 Evaluation

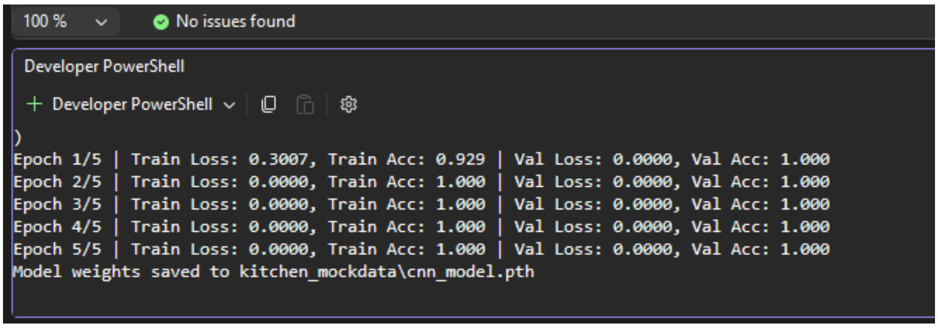

Figure 13: Training output with epochs, loss, accuracy

Figure 14: Model Evaluation on Sample Test Image

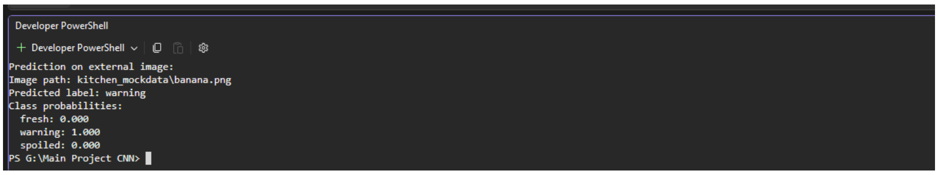

Figure 15: Prediction on external image

For evaluation, training performances, validation results, and output produced by the CNN model are carried out. The goal of this evaluation is to see whether the pipeline is working accurately and also whether the model is perfectly classifying food items on the basis of spoilage level.

The system successfully generated 600 food images(synthetic). The model is trained using five epochs as shown in 13.While observing, it is found that our model achieved an accuracy of 90% in the first epoch and 100% after the second. In the same way, the validation accuracy achieved and remained constant at 100 %. The loss (training and validation) value is nearly 0 after the first epoch, indicating that the CNN model has effectively learned the visual pattern. After the process of training, the model is evaluated using a random image. The output shows a warning image with a prediction confidence of 1.00, while probabilities for fresh and spoiled are 0.000 as shown in 14. This shows the model can make an accurate prediction.

Finally, the trained CNN model is being evaluated using an external image. The image was a yellow banana with some dark spots(banana.png). The model located the image, ran it through the pipeline, and generated the required output. The model showed it belongs to the category "Warning" with probability 1.000 as shown in 15 because at the time of training, yellow colored food items with an average number of spots were associated with the label warning. This prediction confirms successful interpretation of external images and somehow ensures functionality with realistic sample input images.

8 Conclusion

This project presented a design and conceptual validation of a Smart Kitchen Monitoring System, mainly focused on enhancing food safety, food waste minimization, and transparency of operations of the restaurant kitchen. The main objective was to research the possibility of combining Industry 4.0 technologies mainly IoT, Artificial Intelligence, Cloud Computing, Digital Twin concepts, and Blockchain to track the food freshness, spoilage cases, and usage of expired ingredients.

The system architecture illustrates how real time sensor data, visual inspection with CNN-based image classification, and an RFID-based expiry validation system can be combined to support smart decision-making in commercial kitchens. The AI-based spoilage detection model, which was trained on synthetic image and sensor data, was able to correctly categorize food items into fresh, warning, and spoiled ones, which confirms the existence of the proposed analytical pipeline. This is a direct response to the research objective of spoilage detection and risk identification in the early stage.

Moreover, the conceptual use of cloud and blockchain technologies meets the goal of providing data integrity, traceability, and transparency. The system will address the focus of the research on trust and accountability in food handling by recording significant incidents like expiry violations and spoilage notifications on an immutable blockchain ledger, which will reduce fraudulent activities and enable regulatory adherence.

Although the system was not physically deployed, the use of mock data and simulated environments make it possible to control the system behavior and determine that the proposed solution was technically feasible. Altogether, the project demonstrate that AI-based and intelligent monitoring system in a kitchen can considerably improve the process of food safety control operation and provide a good basis to this task in the real world and further implementation.

9 Future work

As we have already explained everything about our project, we still feel that we have some limitations. According to Section 5.4 Two key limitations among those are limited real sensor data and no deployment in the restaurant. To work with this limitation, we created our own mock data to feed and train our model. Also, we have to create our own mock images for the CNN camera model. In the future, we aim to address these two limitations to enhance the quality of our project. We want to deploy our smart kitchen monitoring system to a restaurant to collect real-time sensor and image data. This would allow our model to undergo robust training and validation of the spoiled food detection and help analyze their performance under practical conditions.

Another promising improvement we want to make in the future is the implementation of a complete blockchain prototype with a smart contract to automatically enforce expiry data and audit trails. And finally, we want to build a user-centric evaluation model with a more minimalist dashboard design with a minimum false alarm rate.

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Appendix Here is the appendix